

Task 1.1

Interview Transcript – iRE Agent (Domain Expert)

Interviewee: iRE Agent (Interactive Requirements Elicitation Agent)

Interviewer: Group01 (TDT4242)

Course: TDT4242 – Software Engineering

Context: Requirements elicitation for *AI Guidebook for Students*

Mode: Text-based interview

Date: 16. January

Question 1

Interviewer:

From the perspective of an academic institution, what problems does the AI Guidebook aim to address for students regarding the use of generative AI?

iRE Agent:

The AI Guidebook aims to address several key problems for students within an academic institution. First, it provides clarity on how to use generative AI tools responsibly, ensuring that students adhere to academic integrity policies. Second, it helps mitigate the risk of unintentional plagiarism or misrepresentation by establishing guidelines for AI usage. Third, it fosters an understanding of ethical considerations surrounding AI, promoting responsible engagement with technology. Lastly, the application supports transparency and traceability in AI usage, enabling institutions to monitor compliance with academic standards.

Question 1.1

Interviewer:

Which of these problems are currently the most critical, and why do they require institutional intervention?

iRE Agent:

The most critical problems are the risk of unintentional plagiarism and the lack of clarity around responsible AI use. Institutional intervention is necessary because these issues directly affect academic integrity, which is fundamental to educational credibility. Without clear guidance, students may unintentionally violate policies and face serious academic

consequences. Ethical awareness also requires a structured, institution-led approach to ensure consistent understanding across the student body.

Question 1.2

Interviewer:

How do these challenges affect academic integrity, learning outcomes, and trust between students and the institution?

iRE Agent:

These challenges undermine academic integrity by increasing the likelihood of plagiarism and misrepresentation, which may result in disciplinary actions affecting students' academic records. Learning outcomes may suffer if students rely too heavily on AI tools instead of developing critical thinking and problem-solving skills. This uncertainty can erode trust between students and the institution if students feel unsupported or confused about acceptable practices. Addressing these challenges is essential to maintaining credibility and fostering a supportive learning environment.

Question 2

Interviewer:

How do students today typically find out whether the use of AI is allowed or prohibited in a specific course or assignment?

iRE Agent:

Students usually consult course syllabi, institutional guidelines, or communicate directly with instructors. However, this information is often unclear or inconsistently presented. Many students also rely on informal sources such as peers or online forums, which can lead to misunderstandings. This highlights the need for a centralized and authoritative resource, such as the AI Guidebook.

Question 2.1

Interviewer:

What shortcomings do you see in how current institutional policies regulate or communicate AI usage?

iRE Agent:

Current policies often lack specificity regarding AI usage and may not be updated frequently enough to reflect rapid technological changes. Communication is sometimes

ineffective, with relevant information buried in long documents or inconsistently distributed. The absence of a centralized and accessible system makes it difficult for students to fully understand and comply with expectations.

Question 3

Interviewer:

What information should students be required to document regarding their AI usage when submitting assignments?

iRE Agent:

Students should document which AI tools were used, the nature of the assistance provided (such as drafting, editing, or research), and how AI contributions were integrated into their work. They should also declare any direct AI-generated outputs included and reflect on ethical considerations and compliance with institutional guidelines. This ensures transparency and accountability.

Question 3.1

Interviewer:

At what level of detail should this documentation exist to remain useful without becoming overly burdensome?

iRE Agent:

The documentation should be concise but informative. A brief summary of one to two sentences per assignment describing the AI tools used, their contributions, and any direct outputs is sufficient. Consistency in documentation format is important to streamline the process.

Question 3.2

Interviewer:

How is AI usage documentation currently handled?

iRE Agent:

Currently, AI usage documentation is not handled systematically. Most institutions lack formal requirements, leading to inconsistent practices across courses. Some instructors encourage informal reflections, but there is no standardized process, which creates confusion and varying expectations.

Question 4

Interviewer:

Which parts of AI usage documentation and compliance checking could realistically be automated today?

iRE Agent:

Automation could handle logging AI tool usage, tracking interactions such as prompts and outputs, and performing rule-based compliance checks against institutional guidelines. Automated usage reports could also be generated to summarize interactions and flag potential issues.

Question 4.1

Interviewer:

What technical, institutional, or legal limitations must be considered when automating these processes?

iRE Agent:

Technical limitations include integration with existing educational systems and ensuring data protection compliance, such as GDPR. Institutional limitations arise from varying policies across departments. Legal considerations include intellectual property rights and ownership of AI-generated content.

Question 4.2

Interviewer:

When during the academic workflow should automated compliance checks be performed, and why?

iRE Agent:

Compliance checks should occur during AI usage logging, at assignment submission, and at the end of the semester. This multi-stage approach enables real-time feedback, ensures submission compliance, and supports institutional review and reporting.

Question 4.3

Interviewer:

What actions should the system take if a potential policy violation is detected?

iRE Agent:

The system should alert the student with details of the issue and guidance on corrective actions. The submission should be flagged for review by academic staff, and educational resources should be provided to help prevent future violations.

Question 5

Interviewer:

What kinds of visualizations would be useful for presenting AI usage data to students?

iRE Agent:

Useful visualizations include usage trends over time, breakdowns of AI tool contributions, assignment-level summaries, and compliance status indicators. These help students understand and reflect on their AI usage patterns.

Question 5.1

Interviewer:

How should these visualizations support student reflection and behavior change?

iRE Agent:

Visualizations should provide actionable insights that help students recognize over-reliance on AI or documentation gaps. Combined with reflective prompts, they can encourage responsible behavior and informed decision-making.

Question 5.2

Interviewer:

Should similar insights be available to instructors or administrators, and under what constraints?

iRE Agent:

Yes, but access should be restricted and data should be aggregated to protect student privacy. Only authorized personnel should have access, and the insights should be used to support learning rather than punitive enforcement.

Question 6

Interviewer:

What ethical, legal, or institutional constraints must the system respect when handling student data?

iRE Agent:

The system must comply with data protection laws such as GDPR or FERPA, ensure informed consent, maintain confidentiality, follow institutional policies, and uphold ethical standards such as fairness and transparency.

Question 6.1

Interviewer:

How should the system balance privacy with transparency and accountability?

iRE Agent:

This balance can be achieved through anonymization, aggregated reporting, clear communication about data usage, strict access controls, and regular audits to safeguard privacy while ensuring accountability.

Question 6.2

Interviewer:

What level of explainability should the system provide when making compliance-related decisions?

iRE Agent:

The system should offer clear explanations for compliance decisions, including which rules were triggered and why. This allows students to understand, learn from, and correct their behavior.

Question 7

Interviewer:

How would you define a successful AI Guidebook?

iRE Agent:

A successful AI Guidebook improves student understanding of responsible AI use, provides intuitive and effective functionality, and promotes ethical integration of AI as a support tool rather than a replacement for learning.

Question 7.1

Interviewer:

Are there measurable KPIs the system should track, and how should they be used?

iRE Agent:

Relevant KPIs include compliance rates, AI tool usage frequency, incident reports, and user

engagement metrics. Institutions can use these to evaluate system effectiveness, adjust policies, and improve student support.

Question 7.2

Interviewer:

What risks or unintended consequences should be avoided?

iRE Agent:

Risks include over-reliance on AI, privacy breaches, misinterpretation of compliance data, unequal access, and resistance to adoption if the system is perceived as punitive. These risks must be mitigated through careful design and ongoing stakeholder involvement.

Question 8

Interviewer

How do you (roughly) imagine the website to look?

iRE Agent

I cannot provide a visual concept for the website. However, I can help outline functional requirements that focus on user interface elements. The system shall include a personal dashboard for usage visualization, clear sections for ethical guidelines, AI usage logging, and assignment-level declarations. Additionally, usability should be prioritized to ensure students can navigate easily and access information without confusion. If you have specific design elements in mind, we can explore their functional implications.

Question 8.1

Interviewer

What pages should the website be divided into?

iRE Agent

The website should be divided into the following key sections:

1. ****Home Page****: An overview of the application and its purpose.
2. ****Dashboard****: A personal area for students to visualize their AI usage patterns.
3. ****AI Usage Logging****: A section for students to log their AI tool usage.
4. ****Assignment-Level Declarations****: A page dedicated to declaring AI assistance for specific assignments.

5. **Ethical Guidelines**: Clear information on ethical use of AI tools and academic integrity.
6. **Compliance Checks**: Automated feedback on adherence to institutional policies.
7. **Help/Support**: Resources and FAQs for user assistance.

Each section should be designed with usability in mind to ensure easy navigation and accessibility of information.

Question 8.2

Interviewer:

What should be on the “home page”?

iRE Agent:

The home page should include:

1. **Introduction**: A brief overview of the AI Guidebook's purpose and benefits for students.
2. **Key Features**: Highlight main functionalities, such as the personal dashboard, usage logging, and ethical guidelines.
3. **Navigation Menu**: Clear links to all major sections of the website for easy access.
4. **Latest Updates**: Announcements or news related to AI tools and academic integrity.
5. **Call to Action**: Encouragement for students to register or log in to start using the application.

This layout will ensure that visitors quickly understand the platform's value and how to navigate it effectively.

Question 9

Interviewer:

How much load should the website be able to handle?

iRE Agent

The website should be designed to handle concurrent users based on expected student enrollment and usage patterns. A preliminary estimate might suggest supporting at least 500 concurrent users during peak hours, considering the scale of student engagement. Additionally, performance metrics should focus on response times under load, aiming for page load times of less than 2 seconds. It's essential to conduct load testing to refine these estimates and ensure robust performance under varying conditions.

Question 9.1

Interviewer:

How much downtime is acceptable?

iRE Agent:

The acceptable downtime for the website should ideally be minimal, targeting a maximum of 1-2 hours per month for maintenance and updates. This translates to a high availability goal of 99.9% uptime, ensuring that students have reliable access to the AI Guidebook at all times. Any extended downtime should be communicated in advance, with contingency plans in place to minimize impact on users.

Interview Transcript – Student

Interviewee: Student

Interviewer: Group01 (TDT4242)

Course: TDT4242 – Software Engineering

Context: Requirements elicitation for *AI Guidebook for Students*

Mode: Text-based interview

Date: 16. January

Question 1

Interviewer:

How would you use generative AI tools in your studies? (chatgpt, copilot etc)

Student:

I use it for a lot of stuff. But mainly for improving texts I write and also help with **coding, for**

example if there is an issue that is hard to resolve. It's also helpful for repetitive tasks, like reformatting text.

Question 1.1

Interviewer:

What do you usually use this AI for mostly? Exercises, learning or other?

Student

Mostly exercises I would say. For learning I try to rely more on written sources, like lecture slides, books, etc. But it can be useful to ask AI to explain a concept if I find it hard to understand.

Question 1.2

Interviewer:

How often would you say that you use AI during a typical week?

Student:

Almost every day. But not necessarily very much each day. There's more usage in periods when there are deliveries etc.

Question 2

Interviewer:

When you use generative AI for academic work, what are you usually trying to achieve?

Student:

I would say I use it for the most part to improve efficiency in my work. This could be using it to find an answer or explanation instead of searching for it using other normal search engines like Google, or reformatting and structuring text which also saves a lot of time.

Question 2.1

Interviewer:

How do you personally balance using AI to save time versus using it to support learning?

Student:

I at least try to understand stuff on my own at first. So I don't ask it right away to tell me how I should solve a problem. But I'll use it if I get stuck or if there are some repetitive tasks that has to be done.

Interviewer:

Can you describe a situation where AI genuinely helped you understand something better?

Student:

I had a course on information retrieval where I didn't quite understand a certain method used. I then gave ChatGPT the information it needed and asked questions to make it explain it. I also gave follow up questions to further gain an understanding about the subject.

Question 3

Interviewer:

Can you walk me through your typical workflow when you use AI for an assignment, from start to finish?

Student:

I'll read the assignment start working on it. I try to not rely on AI initially as I think it can be a bad habit to start asking the AI without even trying to think about a solution yourself. The workflow after that depends a bit on what type of assignment it is. If the assignment is a text, then I'll probably write a bit and then ask the AI to improve a paragraph or something when I feel like my writing could be improved or I can't think of a fitting word or something. I might also write a bit sloppy just to get the idea down, and then ask AI to express it a bit more formal. If it's coding, I'll usually ask it for help if I have an error I can't figure out. I might also "discuss" a bit with it if a certain approach would be good before going ahead with an idea.

Question 3.1

Interviewer

At what stages of the workflow do you feel AI is most useful?

Student:

I think AI is most useful in the middle of and in the end of the work. At the beginning of the work I think it's better to try to think for yourself.

Question 3.2

Interviewer:

Are there stages where you feel unsure or uncomfortable about whether AI use is allowed?

Student:

Not really. If the teacher haven't said anything, then I'll consider it to be allowed to be used for the entirety of the project. To a certain degree of course. Not generating an entire project.

Question 4

Interviewer:

How do you usually determine whether using AI is allowed for a specific course or assignment?

- a. Have you ever been uncertain about whether your AI usage might violate academic integrity rules?
- b. What do you typically do when the rules around AI usage are unclear?

Student:

- a. Usually I find out from the teacher saying specifically. I have noticed that they have started explicitly saying that they want us to use AI to learn for certain courses. But not all teachers say anything about AI usage.
- b. If no rules are set concerning AI usage, I just use it however I would like myself. If they are unclear or I don't understand the rules I would ask my peers or the teacher.

Question 5

Interviewer:

What challenges or frustrations have you experienced when using AI tools for academic work?

Student

It's mostly a useful tool I think. One issue could be overreliance. It's can be a bit too easy to ask AI. Without even trying to figure out things yourself.

Question 5.1**Interviewer:**

Have you ever worried about accidentally violating AI-related rules or policies? (if 4.1 was to vague of an answer ask this)

Student:

Not really.

Question 5.2**Interviewer:**

Can you describe a specific situation where AI usage caused confusion or stress?

Student:

Not much. But maybe if we in a group project get into the habit of using AI for everything. That can be a bit bad.

Question 6**Interviewer:**

Have you gotten an AI declaration that you needed to fill out for a delivery/assignment?

https://i.ntnu.no/documents/portlet_file_entry/1305837853/NO_KI+-+deklareringsskjema+per+v%C3%A5r+2024.pdf/3b07aee3-d348-4360-e595-057ec480f1a3 (show this if not)

Student:

Yes.

Question 6.1

Interviewer:

What kind of information about your AI usage would feel reasonable and what do you feel like is unnecessary?

Student:

From my experience, all the AI declarations that I have filled out have asked about reasonable things about my AI usage.

Question 6.2

Interviewer:

What would make documentation feel useful rather than like extra administrative work?

Student:

I'm not sure any documentation would be useful for me, but I can image that declaration or maybe logging could be useful for schools that integrate AI into the learning process.

Question 7

Interviewer:

If NTNU introduced an AI Guidebook to help students use AI responsibly, what would you expect from such a system? (explain what a guidebook in this example may mean but be vague)

Student:

I would expect it to be a place where you could easily find what the policies are for each subject/assignment. Also make it easier to document usage, perhaps make it automatic in some way. It can be really tiresome to do manually.

Question 7.1

Interviewer:

Which features would be most valuable to you as a student?

Student:

What I said in my previous answer.

Question 7.2

Interviewer:

Are there any features or behaviors you would not want from such a system?

Student:

I would not like it to be overly intrusive.

Question 8

Interviewer:

What kinds of visualizations or feedback about your AI usage would you find interesting or helpful?

Student:

I think it would be interesting and useful to get feedback on my prompting or questions, specifically when using AI for coding support.

Question 8.1

Interviewer:

Could you use this information to adjust your study habits or AI usage?

Student:

Yes, if I got feedback on how to improve or better use AI with coding, it would improve my efficiency even more.

Question 8.2

Interviewer:

Do you think seeing trends over time influence how you use AI? Why or why not?

Student:

I don't think so but it would depend on the specific trend. If it would show for example bad habits over time, then this could help me improve as well.

Question 9

Interviewer:

Do you have any ethical or privacy concerns related to using AI tools in your studies?

Student:

Not really. There are obvious things like not sharing personal/sensitive data with AI, but I feel like that is not specific to studies. Stating that images are AI generated perhaps.

Question 9.1

Interviewer:

How would you feel about a system that logs or analyzes your AI usage?

Student:

It's a bit creepy if it analyzes everything of AI usage on my computer. But if it only analyzes for school work etc, when using the system, it's fine.

Question 9.2

Interviewer

Is there anything that would make you trust such a system?

Student

If it clearly states what data is collected. And how it uses it, saves it, etc.

Question 10

Interviewer:

What would motivate you to actually read and follow AI usage guidelines?

Student:

If a course had strict rules for AI usage for assignments or projects, I would feel obliged to read the guidelines.

Question 10.1

Interviewer:

Is there anything that makes you ignore guidelines or policy documents?

Student:

Long texts of guidelines or policies is not something I would read, at least not in full detail.

Question 10.2

Interviewer:

Is there a way guidelines can be presented to make them more useful or engaging for you?

Student:

Maybe through dynamic feedback from usage of AI. If they had to be presented beforehand, maybe with clear and as few points as possible so it's not tedious to read.

Functional requirements:

FR1

As a student, I want to access clear and up-to-date AI usage policies for each course or assignment in one place, so that I can understand what is allowed and avoid unintentionally violating academic integrity rules. (ire question 2, student question 4, 7)

FR2

As a student, I want to document which AI tools I used and how they contributed to an assignment, so that my work is transparent and compliant with institutional requirements. (ire question 3, 3.1, 8, 8.1, student question 6)

FR3

As a student, I want AI usage documentation to be concise and structured, so that declaring my AI use does not feel like unnecessary administrative overhead. (ire question 3.1, student question 6, 10)

FR4

As a student, I want the AI Guidebook to automatically log the AI tool used, timestamp, assignment context, and type of assistance (e.g., text editing, code explanation, idea generation) when I interact with AI through the system, so that I do not have to manually remember and report this information during assignment submission. (ire question 4, student question 7)

FR5

As a student, I want the system to check my AI usage for policy compliance during logging and assignment submission, so that I receive early feedback and can correct issues before final delivery. (ire question 4.2, 4.3)

FR6

As a student, I want clear explanations when my AI usage is flagged as potentially non-compliant, so that I understand what rule was triggered and how to fix the issue. (ire question 6.2, 8.1, student question 9.2)

FR7

As a student, I want access to ethical guidelines and learning-oriented feedback when using AI, so that AI supports my learning instead of replacing my own reasoning. (ire question 1, 7, student question 2)

FR8

As a student, I want to see visual summaries such as line charts showing AI usage frequency over time, bar charts comparing AI usage per assignment, and compliance status indicators, so that I can reflect on my usage patterns and avoid over-reliance on AI tools. (ire question 5, 8, 8.1, student question 8)

FR9

As an instructor or administrator, I want access to aggregated AI usage insights across courses, so that I can identify trends and support students without accessing individual-level data. (ire question 5.2)

FR10

As an academic institution, I want potential AI policy violations to be flagged and reviewed by staff rather than automatically penalized, so that academic integrity is upheld while maintaining a supportive learning environment. (ire question 4.3, 7.2)

FR11

As a student, I want to access a home page that explains the purpose of the application, so that I can quickly understand what the system is for. (ire question 8.1, 8.2)

FR12

As a student, I want to see the key features of the system on the home page, so that I understand what functionalities are available to me. (ire question 8.2)

FR13

As a student, I want a clear navigation menu on the home page, so that I can easily access all major sections of the website. (ire question 8.2)

FR14

As a student, I want to view announcements or updates related to AI tools and academic integrity, so that I stay informed about relevant changes. (ire question 8.2)

FR15

As a visitor, I want be able to register or log in on the home page, so that I can start using the application. (ire question 8.2)

FR16

As a user, I want a help and support section with FAQs and resources, so that I can resolve issues without external assistance (ire question 8.1)

FR17

As a student, I want to be informed in advance of any planned downtime, so that I can plan my work accordingly and avoid unexpected interruptions. (ire question 9)

Non-functional requirements:

NFR1

As a student, I want my AI usage data to be stored and processed in accordance with GDPR requirements, including data minimization, access control, and the ability to delete my data upon request, so that my personal information and academic work remain secure. (ire question 6)

NFR2

As a student, I want the system to present a dedicated “Data Usage” page that explicitly lists what AI usage data is collected, how long it is stored, and who has access to it, so that I can make informed decisions about using the system. (ire question 6.1, student question 9.2)

NFR3

As a student, I want each compliance warning to include the specific policy rule triggered, a short explanation of why it was triggered, and suggested corrective actions, so that I can understand and improve my AI usage. (ire question 6.2, student question 10)

NFR4

As a student, I want AI usage logging and compliance feedback to occur automatically in the background without interrupting my primary task flow, except when explicit user action or confirmation is required, so that the system supports my studies rather than disrupting them. (student question 7.2)

NFR5

As an academic institution, I want the system to support course-specific AI usage rules that can be configured per course or assignment, so that different departments can enforce their own policies without modifying the core system. (ire question 4.1)

NFR6

As an academic institution, I want compliance feedback to prioritize educational messages and guidance before escalation to staff review, so that students experience the system as supportive rather than punitive. (ire question 7.2)

NFR7

As a student, I want all primary system functions (dashboard, logging, declarations, guidelines, support) to be accessible within a maximum of two navigation steps from the home page, so that information is easy to locate. (ire question 8)

NFR8

As an academic institution, I want the website to support at least 500 concurrent users during peak usage periods, so that all students can access the system simultaneously without degradation of service. (ire question 9)

NFR9

As a student, I want pages on the website to load within 2 seconds under normal and peak load conditions, so that I can use the system efficiently without delays. (ire question 9)

NFR10

As an academic institution, I want the system to achieve at least 99.9% uptime, so that students have reliable access to the platform throughout the academic term. (ire question 9.1)

NFR11

As a student, I want system downtime to be limited to a maximum of 1 hour per month, excluding scheduled maintenance announced at least 48 hours in advance, so that my access to the AI Guidebook is rarely disrupted. (ire question 9.1)

Task 1.2

AI Usage Logging

FR-1.1: Manual Usage Entry

Defect: Ambiguity

Issue: Unclear whether all the parts should be in separate input fields or one big free-text field.

Impact: Developers does not implement the form as expected by stakeholders.

Defect: Incompleteness

Issue: No information about what type of input fields these are. Are they all free-text fields or are some drop-downs?

Impact: Developers does not implement the form as expected by stakeholders.

FR-1.2: Automatic Tool Integration

Defect: Ambiguity

Issue: “Common AI tools”. Are the ones in parenthesis only examples, or is it these four tools that should be supported? Are there other common tools that should be supported?

Impact: Developer is unsure which AI tools to implement support for.

Defect: Ambiguity

Issue: “Usage frequency”. What is this number? Number of prompts, sessions, something else...?

Impact: Developer implements an unexpected frequency metric.

Defect: Ambiguity

Issue: “Duration”. What is this number? Time while prompting, while having the tool open, something else...?

Impact: Developer implements an unexpected duration metric.

FR-1.3: Usage Categorization

Defect: Incompleteness

Issue: Unclear if it should be possible to categorize the AI usage by more than one task type.

Impact: Developer has to make assumptions.

Defect: Inconsistency/incompleteness

Issue: FR-1.1 states that “context” should be logged. It’s unclear what context is, but it can be interpreted as what is described as “categorization” in this requirement. If so, the same

object is described with different terms. If they are two different things, then should the category be registered at a different time than the other things in FR-1.1?

Impact: Confusion for the developer, unsure if FR-1.1 and FR-1.3 describes the same thing or not.

AI Declaration Management

FR-2.1: Assignment-Level Declaration

Defect: None

Assessment: Its well defined, complete and understandable

FR-2.2: Declaration Amendment

Defect: Incompleteness

Issue: Does not specify whether amendments trigger re-evaluation or notifications

Impact: Unclear compliance implications

FR-2.3: Pre-Assignment Consultation

Defect: Ambiguity / Unverifiable requirement

Issue: Students can “request clarification” before assignment submission, but it does not clearly define what system behavior constitutes a fulfilled request

Impact: Cannot be objectively tested or validated, as successful fulfillment relies on instructor actions outside system control

Student Dashboard and Visualization

FR- 3.1: Usage Overview Dashboard

Defect: Ambiguity / Incomplete

Issue: It is unclear what the tool is in “Frequency by tool”, sessions, prompts, minutes? It is incomplete because it is not specified what time range is meant by “over time”, and whether this can be changed by the user.

Impact: Because it is ambiguous the wrong frequency by tool might be implemented. And because it is incomplete, expected features surrounding the time range might not be implemented.

FR- 3.2: Compliance Status Display

Defect: Unverifiability

Issue: QA cannot verify compliance logic without explicit rule definitions.

Impact: Without defined rules, you can't write test cases or acceptance criteria.

FR- 3.3: Comparative Analytics

Defect: Ambiguity

Issue: It is not clear what is meant by "frequency"? Sessions, prompts?

Impact: Wrong frequency might be implemented.

FR-4.1: Aggregated Usage Reports

Defect: Ambiguity

Issue: "Overall AI usage trends". What metrics are these and how many of them should there be? It seems like it's more than one, as "trends" is in plural, but we do not know how many.

Impact: Developer does now know what/how many "trends" to implement.

FR-4.2: Individual Declaration Review

Defect: Ambiguity/Inconsistency

Issue: "AI declarations". Is this the same as "AI tool usage" in FR-1.1, or is it something else? If it's something else, then it needs to be stated. If it's the same, then there is an inconsistency in terms.

Impact: Developer does not know what to implement and is confused about naming.

FR-4.3: Anomaly Notifications

Defect: Incompleteness

Issue: “Unusual usage patterns” - what is considered unusual? “Threshold violations” - what is this threshold? Will it be defined by the teacher somewhere or is it hardcoded in the system?

Impact: Developers do not know the conditions for what should trigger flagging.

Defect: Verifiability

Issue: “Identify potential compliance issues early”. What is considered early? How do you measure this?

Impact: Can not really test the requirement.

FR- 4.4

Defect: Ambiguity

Issue: “Communicate directly”. One could assume this is a real-time chat. But it could also be a comment thread, mail system, etc.

Impact: Developers may implement a communication system that is not expected.

Academic Integrity Officer Functions

FR-5.1: Case Management System

Defect: Ambiguity / Privacy inconsistency

Issue: “detailed logs, student declarations, and incident history” but does not specify access constraints, data scope, or anonymization rules

Impact: Unclear access boundaries

FR-5.2: Violation Documentation

Defect: Incompleteness

Issue: Does not define where or how investigation outcomes and communications are stored, or whether documentation is versioned, auditable, or access restricted

Impact: Lack of specification

FR-5.3: Advanced Analytics

Defect: Ambiguity

Issue: The terms “analyze trends” and “identify systemic issues” are not operationally defined

Impact: Cannot be objectively verified or tested

University Administrator Functions

FR- 6.1: Institutional Reporting

Defect: Incompleteness

Issue: Requirement allows access to institutional analytics without defining authorization controls, auditability, or aggregation safeguards.

Impact: Could lead to unauthorized access, privacy breaches, and non-compliance with institutional data governance policies.

FR- 6.2: Policy Management Interface

Defect: Incompleteness

Issue: The requirement does not specify critical details needed to implement the policy management interface, such as policy format (text vs structured rules), versioning, approval/publishing workflow, and access control/audit logging for policy changes.

Impact: This can lead to unclear scope and inconsistent implementations, and it creates security and governance risks (e.g., unauthorized policy changes or lack of traceability), making the feature difficult to test and accept.

FR- 7.1

Defect: Ambiguity

Issue: “Access the AI Guidebook system through my university’s learning management system”. This has different interpretation. Is data fetched by an API to Blackboard, is it in a frame integrated into Blackboard, or is it just a redirect link to the Guidebook?

Impact: Developer implements an unexpected integration.

Defect: Incompleteness

Issue: Which learning management system should be supported? I imagine it is hard to support everything. Is it Blackboard, Canvas... something else?

Impact: Very big, probably impossible, scope for the developer.

Defect: Incompleteness

Issue: Authentication behavior is not specified. Do the users need to reauthenticate to get to the Guidebook, or does it use the same sign-in as the learning management system?

Impact: Developer does not know what kind of authentication system to implement.

Defect: Verifiability

Issue: “Seamlessly integrate AI documentation into my existing workflow”. This is subjective and hard to measure. Should be more specific what it means for it to be seamless.

Impact: Can not test the requirement.

Usability Requirements

NFR-1: Learnability

Defect: None

Assessment: Quantifiable, testable

NFR- 2: Task Efficiency

Defect: Inconsistency

Issue: This requirement talks about a pre-filled template. FR-1.1 on the other hand talks about manual logging the AI tool usage.

Impact: It's unclear to the developer whether the form should be empty (for manual filling) or if there are pre-filled templates.

NFR-3: Mobile Operability

Defect: Ambiguity

Issue: The requirement uses the term “all core system functions”, which is not precisely defined. Even though examples are listed (manual logging, declaration submission, dashboard viewing), it is unclear whether these are the complete set of “core” functions or just examples.

Impact: Different developers/testers may interpret the scope differently, leading to gaps in mobile support (features missing or partially working on mobile). This can also cause disagreements during acceptance testing because the team may believe the requirement is met while stakeholders expect additional functionality to be included.

Performance Requirements

NFR-4: Response Time

Defect: None

Assessment: Quantifiable, testable

NFR-5: System Availability

Defect: None

Assessment: Clear and verifiable

Privacy and Security Requirements

NFR-6: Prompt Data Isolation

Defect: None

Assessment: Clear and verifiable

NFR-7: Data Encryption

Defect: None

Assessment: Clear, consistent, and verifiable

NFR-8: Minimal Data Collection

Defect: Inconsistency

Issue: The requirement says only tool name, usage duration, task category, assignment ID, timestamp should be collected. FR-1.1 on the other hand says something called “context” also should be collected.

Impact: More, or less, data is collected than the stakeholders expect.

Defect: Scope Ambiguity

Issue: “During normal operation”. Does this mean more personal data can be collected during non-normal operation. If so, which data is this?

Impact: Data collected under exceptional or administrative processes may fall into a grey area.

NFR-9: AI Tool Integration

Defect: Unverifiable

Issue: The requirement states that a developer should be able to integrate a new AI tool “in under 10 hours without modifying core system code”, this is nonverifiable. The “under 10 hours” metric is subjective, depending on developer experience, familiarity, and environment. There is no finite, repeatable process to test this for all developers.

Impact: QA cannot reliably test whether the requirement is met, making acceptance decisions subjective. Could create unrealistic expectations about system extensibility and maintainability.

Task 1.3

```
{
  "FR1": {
    "depends_on": [],
    "enables": ["FR2", "FR5", "FR7"],
    "conflicts_with": [],
    "refines": [],
    "duplicates": [],
    "overlap": ["NFR5"]
  },
  "FR2": {
    "depends_on": ["FR1"],
    "enables": ["FR3", "FR4", "FR5"],
    "conflicts_with": [],
    "refines": [],
    "duplicates": [],
```

```
"overlap": []
},
"FR3": {
  "depends_on": ["FR2"],
  "enables": [],
  "conflicts_with": [],
  "refines": ["FR2"],
  "duplicates": [],
  "overlap": []
},
"FR4": {
  "depends_on": ["FR2", "NFR1", "NFR4"],
  "enables": ["FR5", "FR8", "FR9"],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": []
},
"FR5": {
  "depends_on": ["FR1", "FR4", "NFR5"],
  "enables": ["FR6", "FR10"],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": ["NFR3"]
},
```

```
"FR6": {
  "depends_on": ["FR5"],
  "enables": [],
  "conflicts_with": [],
  "refines": ["FR5"],
  "duplicates": [],
  "overlap": ["NFR3"]
},
"FR7": {
  "depends_on": ["FR1"],
  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": ["NFR6"]
},
"FR8": {
  "depends_on": ["FR4"],
  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": ["FR9"]
},
"FR9": {
  "depends_on": ["FR4", "NFR1"],
```



```
"enables": [],
"conflicts_with": [],
"refines": [],
"duplicates": [],
"overlap": ["FR8"]
},
"FR10": {
  "depends_on": ["FR5"],
  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": ["NFR6"]
},
"FR11": {
  "depends_on": [],
  "enables": ["FR12", "FR13", "FR15"],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": ["NFR7"]
},
"FR12": {
  "depends_on": ["FR11"],
  "enables": [],
  "conflicts_with": [],
```

```
"refines": ["FR11"],
"duplicates": [],
"overlap": []
},
"FR13": {
  "depends_on": ["FR11"],
  "enables": ["FR16"],
  "conflicts_with": [],
  "refines": ["FR11"],
  "duplicates": [],
  "overlap": ["NFR7"]
},
"FR14": {
  "depends_on": ["FR11"],
  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": ["FR17"]
},
"FR15": {
  "depends_on": ["FR11"],
  "enables": ["FR2"],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
```

```
"overlap": []  
},  
"FR16": {  
  "depends_on": ["FR13"],  
  "enables": [],  
  "conflicts_with": [],  
  "refines": [],  
  "duplicates": [],  
  "overlap": []  
},  
"FR17": {  
  "depends_on": ["NFR10", "NFR11"],  
  "enables": [],  
  "conflicts_with": [],  
  "refines": [],  
  "duplicates": [],  
  "overlap": []  
},
```

```
"NFR1": {  
  "depends_on": [],  
  "enables": ["FR4", "FR9"],  
  "conflicts_with": [],  
  "refines": [],  
  "duplicates": [],  
  "overlap": []
```

```
},  
"NFR2": {  
  "depends_on": ["NFR1"],  
  "enables": [],  
  "conflicts_with": [],  
  "refines": [],  
  "duplicates": [],  
  "overlap": []  
},  
"NFR3": {  
  "depends_on": ["FR5"],  
  "enables": [],  
  "conflicts_with": [],  
  "refines": [],  
  "duplicates": [],  
  "overlap": ["FR6"]  
},  
"NFR4": {  
  "depends_on": [],  
  "enables": ["FR4"],  
  "conflicts_with": [],  
  "refines": [],  
  "duplicates": [],  
  "overlap": []  
},  
"NFR5": {
```

```
"depends_on": [],
"enables": ["FR1", "FR5"],
"conflicts_with": [],
"refines": [],
"duplicates": [],
"overlap": []
},
"NFR6": {
  "depends_on": [],
  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": ["FR7", "FR10"]
},
"NFR7": {
  "depends_on": [],
  "enables": ["FR11", "FR13"],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": []
},
"NFR8": {
  "depends_on": [],
  "enables": [],
```

```
"conflicts_with": [],
"refines": [],
"duplicates": [],
"overlap": []
},
"NFR9": {
  "depends_on": ["NFR8"],
  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": []
},
"NFR10": {
  "depends_on": [],
  "enables": ["FR17"],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlap": []
},
"NFR11": {
  "depends_on": ["NFR10"],
  "enables": ["FR17"],
  "conflicts_with": [],
  "refines": [],
```

```
"duplicates": [],  
"overlap": []  
}  
}
```

Requirements related to system entry and orientation (FR11 to FR16) can be implemented first because they have no functional dependencies and provide users with access, navigation, and support. In the same initial phase, FR1 (access to AI usage policies) is implemented in parallel, as it supplies essential policy context required for later documentation, compliance checking, and ethical guidance, even though it does not technically depend on the home page features.

Once this is done, functionality for documenting and logging AI usage (FR2 to FR4) can be implemented. These requirements depend on both policy availability and non-functional constraints such as data protection (NFR1) and non-intrusive background operation (NFR4). Compliance-related features (FR5 and FR6) are after, as they rely on the existence of logged usage data and course-specific policy rules (NFR5). Ethical and educational guidance (FR7) and staff review processes (FR10) are implemented alongside compliance checks, with their behavior shaped by non-punitive non-functional requirements (NFR6).

Then, analytical features for students and instructors (FR8 and FR9) can be implemented after the core data pipeline is stable and privacy constraints are satisfied. Availability related functionality such as downtime notifications (FR17) is fixed during deployment and operational planning, influenced by uptime and maintenance requirements (NFR10 and NFR11).